

- **Module 14) Python – Collections, functions and Modules**

4. Tuple

- **Introduction to tuples, immutability.**

A tuple is an ordered, immutable sequence data type in Python. It is similar to a list but cannot be modified after creation.

Characteristics of Tuples:

- **Ordered:** Elements maintain their position (index-based access).
- **Immutable:** Once created, elements cannot be added, removed, or changed.
- **Heterogeneous:** Can store elements of different data types.
- **Hashable (if all elements are hashable):** Can be used as keys in dictionaries.
- **Memory Efficient:** Typically consumes less memory than lists due to immutability.

2. Immutability in Tuples

What Does Immutability Mean?

- Once a tuple is created, its **structure and elements cannot be altered**.
- No methods exist to modify the tuple (unlike lists, which have `append()`, `remove()`, etc.).
- Attempting to modify a tuple raises a `TypeError`.

- **Creating and accessing elements in a tuple**

Syntax for Creating Tuples

Tuples are defined using **parentheses** () (optional in some cases) with elements separated by commas.

Methods of Creating Tuples:

1. **Standard Tuple**

```
my_tuple = (1, 2, 3, "hello")
```

Single-Element Tuple (Requires a Trailing Comma)

```
single_tuple = (4,) # Without comma, Python treats it as an integer
```

Empty Tuple

```
empty_tuple = ()
```

2. Accessing Tuple Elements

A. Indexing (Zero-Based)

- Tuples support **positive** (left-to-right) and **negative** (right-to-left) indexing.
- **Syntax:** tuple[index]

- **Basic operations with tuples: concatenation, repetition, membership.**

1. Concatenation (+ Operator)

Definition: Combining two or more tuples into a **new tuple**.

Key Properties:

- **Creates a new tuple** (original tuples remain unchanged).
- **Order is preserved** (elements appear in the same sequence).
- **Works only with tuples** (cannot concatenate a tuple with a list directly).

Syntax:

```
new_tuple = tuple1 + tuple2 + tuple3 ...
```

2. Repetition (* Operator)

Definition: Repeating a tuple a specified number of times to create a **new tuple**.

Key Properties:

- **Returns a new tuple** (original remains unchanged).
- **Requires an integer multiplier (n).**
- If $n \leq 0$, returns an **empty tuple**.

Syntax:

```
new_tuple = tuple * n
```

3. Membership Testing (in & not in Operators)

Definition: Checking if an element exists in a tuple.

Key Properties:

- Returns True **or** False.
- Performs **linear search** (checks each element sequentially).
- **Case-sensitive** for strings.
- Works with **any data type** (checks using `==` comparison).

- **Syntax:**

element in tuple

element not in tuple